TrafficSense — Multi-Agent Smart City Traffic Management System

Capstone Project | School of Computer Science & Engineering | VIT University

■ Student Information

Name	Darsi Amit
Register No.	23BCE8288
Programme	B.Tech — Computer Science and Engineering
University	VIT-AP University, Amaravati
Project Title	TrafficSense: A Multi-Agent Smart City Traffic Management System
Role	Frontend & Dashboard Lead — responsible for the complete Streamlit multi-page dashboard, CSS theming, live detection UI, and Plotly analytics visualisations
Academic Year	FALL SEM 2026-27
Faculty Mentor	Dr. Mohd Irfan, Assistant Professor Sr. Grade-1, SCOPE, VIT-AP University

■ Contribution Overview

My primary responsibility was designing and building the complete real-time web dashboard for the TrafficSense system. This includes all four Streamlit pages, the custom CSS glassmorphism theme, interactive Plotly charts, the live video detection interface, and the real-time agent cooperation visualisation. The dashboard serves as the primary interface for city planners and faculty to interact with and demonstrate the system.

■ Detailed Technical Contributions

1. Dashboard Architecture & Theming

- Designed and implemented the multi-page Streamlit dashboard using Streamlit's native Page navigation API.
- Created a comprehensive glassmorphism CSS theme (src/dashboard/utils.py) with Google Inter fonts, gradient buttons, pulsing animations, and glowing metric cards.

• Implemented st.html() rendering to inject complex HTML/CSS components without Streamlit's markdown parser interfering.

• Built reusable component library: metric_card.py and intersection_card.py for consistent UI across pages.

2. Landing Page (app.py)

• Designed the Hero section with animated gradient headline and project tagline.

• Built responsive 4-column metric grid showing live system-wide statistics.

• Created visual navigation cards with icons and descriptions for each dashboard section.

• Added a Model Health status card showing the current AI model's mAP, training date, and active classes.

3. Live Traffic Monitor (pages/1_live_monitor.py)

• Built the complete video upload and real-time YOLO inference interface.

• Implemented a three-mode model selector: Custom Indian (24 classes), COCO Pretrained (80 classes), and YOLO-World (Open Vocabulary) with a live text-input class selector.

• Designed the side-by-side layout: video feed on the left, live telemetry metrics panel on the right.

• Integrated real-time glowing metric cards showing Active Entities, Congestion Level, Avg Confidence, and Processing FPS.

• Built the post-processing analytics section with vehicle class distribution bar chart.

4. Network Control Centre (pages/2_network_control.py)

• Created a 2×2 visual intersection grid representing the physical 4-intersection road network.

• Implemented LED-style glowing dot indicators showing current traffic light phase (Red/Yellow/Green) for each intersection.

• Built the Cooperative Reasoning Log — a live scrolling feed of LLM agent decision messages.

• Designed the Agent Cooperation Map showing message flow between intersections.

5. Analytics Page (pages/3_analytics.py)

• Built a full analytics suite using Plotly with the dark theme.

• Implemented comparative charts: TrafficSense vs. Fixed Timer vs. Random baseline.

• Created the Executive Summary table showing % improvement in wait time, queue length, and throughput.

• Embedded training performance visualisations (confusion matrix, F1 curves) from the YOLOv8 training outputs.

■ Deliverables & Files

File / Module	Description	Status

src/dashboard/app.py	Landing page — Hero, metrics, navigation	✔ Complete
src/dashboard/utils.py	Global CSS theme + helper functions	✔ Complete
src/dashboard/pages/1_live_monitor.py	Live YOLO detection interface	✔ Complete
src/dashboard/pages/2_network_control.py	Intersection grid + agent cooperation log	✔ Complete
src/dashboard/pages/3_analytics.py	Plotly charts + executive summary	✔ Complete
src/dashboard/components/metric_card.py	Reusable glowing metric card component	✔ Complete
src/dashboard/components/intersection_card.py	Intersection LED status card	✔ Complete

■ Technologies & Tools Used

Framework	Streamlit v1.45.1
Charts	Plotly v5.x (dark theme)
Styling	Custom CSS — Glassmorphism, Google Inter Font, CSS animations
HTML Rendering	st.html() native Streamlit API
Video Processing	OpenCV + Streamlit video display
Language	Python 3.12
Port	8505 (localhost)

■ Declaration

I, Darsi Amit (23BCE8288), hereby declare that the contributions described in this report — including the full Streamlit dashboard design, all four pages, CSS theming, component library, and Plotly visualisations — are my individual work carried out as part of the TrafficSense Capstone Project.

Signature: _____ Date: _____